Boundary Layer Meteorology Reference:
Stability Functions Based Upon Shear Functions

Equation Reference Guide (Adapted from England & McNider, 1995)

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1 Exchange Coefficients & Surface Layer Framework

The Monin–Obukhov exchange coefficients for momentum (K_m) and heat (K_h) are defined as:

$$K_m = \frac{l_m u_*}{\phi_m} = \frac{l_m^2 s}{\phi_m^2} \tag{1}$$

$$K_h = \frac{l_m u_*}{\phi_h} = \frac{l_m^2 s}{\phi_m \phi_h} \tag{2}$$

where friction velocity u_*, wind shear s, mixing length l_m, and temperature scale θ_* are:

$$u_* = \frac{l_m s}{\phi_m}, \qquad s = \left[\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2\right]^{1/2} \tag{3}$$

$$l_m = \kappa z, \qquad \theta_* = \frac{l_m}{\phi_h}\frac{\partial \theta}{\partial z} \tag{4}$$

The dimensionless shear functions $\phi(\zeta)$ depend on stability parameter $\zeta = z/L$:

- **Stable branch ($\zeta > 0$):**

$$\phi_m(\zeta) = 1 + \beta_m \zeta, \qquad \phi_h(\zeta) = 1 + \beta_h \zeta \tag{5}$$

- **Unstable branch ($\zeta < 0$):**

$$\phi_m(\zeta) = (1 - b_m \zeta)^{-1/4}, \qquad \phi_h(\zeta) = (1 - b_h \zeta)^{-1/2} \tag{6}$$

Notation Note: Stable constants (β_m, β_h) and unstable constants (b_m, b_h) are treated as distinct empirical parameters.

2 Gradient Richardson Number (Ri) & Inverse Relations

The gradient Richardson number is related to Monin–Obukhov stability ζ via:

$$Ri = \frac{g}{\theta_0}\frac{\partial \theta / \partial z}{s^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} \tag{7}$$

Stable Boundary Layer ($\zeta > 0$)

Substituting stable shear functions into $Ri(\zeta)$ yields the quadratic equation:

$$(Ri\beta_m^2 - \beta_h)\zeta^2 + (2Ri\beta_m - 1)\zeta + Ri = 0 \tag{8}$$

The physically continuous solution requiring $\zeta \to 0$ as $Ri \to 0$ (neutral limit) is:

$$\zeta = \frac{1 - 2Ri\beta_m - \sqrt{1 + 4(\beta_h - \beta_m)Ri}}{2(Ri\beta_m^2 - \beta_h)} \tag{9}$$

- **Critical Richardson Number:**

$$Ri_c = \frac{\beta_h}{\beta_m^2} \tag{10}$$

  This value corresponds to the singular limit of the denominator in the stable inverse relation ($\zeta \to +\infty$).

- **Exact Linear Simplification ($\beta_h = \beta_m = \beta$):** When heat and momentum constants match, the discriminant collapses to $\Delta \equiv 1$ for all Ri, yielding:

$$\zeta = \frac{Ri}{1 - Ri/Ri_c} \qquad \text{where } Ri_c = \frac{1}{\beta} \tag{11}$$

Unstable Boundary Layer ($\zeta < 0$)

Squaring and clearing denominators for the unstable branch yields the cubic equation:

$$b_m\zeta^3 - \zeta^2 - b_h Ri^2 \zeta + Ri^2 = 0 \tag{12}$$

1. Reduction to Depressed Cubic

Dividing by b_m and substituting $\zeta = t + \dfrac{1}{3b_m}$ eliminates the quadratic term, yielding $t^3 + pt + q = 0$ with:

$$p = -\frac{b_h Ri^2}{b_m} - \frac{1}{3b_m^2}, \qquad q = \frac{(3b_m - b_h)Ri^2}{3b_m^2} - \frac{2}{27b_m^3} \tag{13}$$

2. Character of Roots & Trigonometric (Viète) Solution

Because $b_m, b_h > 0$, $p < 0$ strictly holds for all $Ri \neq 0$. This places the system in the *casus irreducibilis* regime (three real roots). The three explicit solutions are:

$$t_k = 2\sqrt{-\frac{p}{3}} \cos\left[\frac{1}{3}\arccos\left(\frac{3q}{2p}\sqrt{\frac{-3}{p}}\right) - \frac{2\pi k}{3}\right], \qquad k = 0, 1, 2 \tag{14}$$

$$\zeta_k = t_k + \frac{1}{3b_m} \tag{15}$$

3. Physical Root Identification

As $Ri \to 0^-$, small-parameter perturbation analysis shows that $-\zeta^2 + Ri^2 \approx 0 \implies \zeta \approx Ri$. The physical branch is uniquely selected by requiring continuity with the neutral limit:

$$\zeta_{\text{physical}} = \min_{k \in \{0,1,2\}} \left\{ \zeta_k : \zeta_k \to Ri \text{ as } Ri \to 0^- \right\} \tag{16}$$

4. Exact Degenerate Factorization ($b_m = b_h = b$)

When momentum and heat parameters coincide, the unstable cubic factors exactly:

$$(b\zeta^3 - \zeta^2 - bRi^2\zeta + Ri^2) = (\zeta - Ri)\left(b\zeta^2 + (bRi - 1)\zeta - Ri\right) = 0 \tag{17}$$

confirming that $\zeta = Ri$ is an exact root in the equal-coefficient limit.

3 Stability Functions $f_m(Ri)$ and $f_h(Ri)$

In numerical models, exchange coefficients are expressed directly in terms of Ri:

$$K_m = f_m(Ri) l_m^2 s, \qquad K_h = f_h(Ri) l_m^2 s \tag{18}$$

where:

$$f_m(Ri) = \phi_m^{-2}(\zeta), \qquad f_h(Ri) = \phi_m^{-1}(\zeta)\phi_h^{-1}(\zeta) \tag{19}$$

Stable Layer Formulations ($0 \leq Ri \leq Ri_c$)

- **Quadratic Form ($Pr = 1$, $\beta_m = \beta_h = \beta$):** Direct substitution of $\zeta = Ri/(1 - \beta Ri)$ into $\phi_m = 1 + \beta\zeta$ yields $\phi_m = (1 - Ri/Ri_c)^{-1}$, proving the quadratic Ri-dependence follows directly from shear functions:

$$f_m(Ri) = f_h(Ri) = \begin{cases} \left(1 - \dfrac{Ri}{Ri_c}\right)^2, & 0 \leq Ri \leq Ri_c \\ 0, & Ri > Ri_c \end{cases} \tag{20}$$

- **Blackadar Linear Form (For Comparison):**

$$f_m(Ri) = f_h(Ri) = \begin{cases} 1 - \dfrac{Ri}{Ri_c}, & 0 \leq Ri \leq Ri_c \\ 0, & Ri > Ri_c \end{cases} \tag{21}$$